MATH 598: Asymptotic Statistics

Fall 2026

Instructor: Mehdi Dagdoug.

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Office: Burnside Hall, 1230.

Lecture time: Monday, Wednesday, from 2:35 pm to 3:55 pm.

Teaching location: Burnside Hall, 1205.

Office hours: Tuesdays, from 1:30 pm to 3:30 pm.

1 Course information

Prerequisites: A first course in probability, such as MATH 323 or MATH 356, and a first course in statistics, such as MATH 324 or MATH 357. A background in analysis would be useful, but is not strictly necessary.

Course description: The course will develop the probabilistic foundations of large-sample theory and applies them to classical and modern problems in statistics. Emphasis will be placed on stochastic convergence, asymptotic expansions, uniform convergence, rates of convergence, and optimality. The final part of the course will introduce minimax rates in nonparametric regression and some semiparametric ideas such as influence functions and orthogonality.
The course will be split into three main parts: (A) Mathematical and probabilistic foundations; (B) Classical large-sample theory; (C) Introduction to minimax and semiparametric theory.

In part (A), the focus will be on mathematical reminders, stochastic convergence, uniform integrability, laws of large numbers, central limit theorems, triangular arrays, and asymptotic linearity.

Part (B) will focus on studying the asymptotic properties of various classical estimators, such as the empirical distribution function and sample quantiles, maximum likelihood, M-estimation, Z-estimation, with various applications to estimation and testing.

Part (C) will be an introduction to selected modern topics such as minimax upper/lower bounds for nonparametric regression and semiparametric theory, as time permits.

Required course materials: None; typed lecture notes will be provided.

Optional course materials: The following textbooks may be useful for additional reading.

- Ferguson, T. S. (2017). *A course in large sample theory*. Routledge.
- van der Vaart, A. W. (2000). *Asymptotic statistics*. Cambridge university press.
- Lehmann, E. L. (1999). *Elements of large-sample theory*. New York, NY: Springer New York.
- Jiang, J. (2010). *Large sample techniques for statistics*. New York: Springer.
- Tsybakov, A. B. (2008). *Introduction to Nonparametric Estimation*. Springer.

Tentative weekly plan: The course will roughly follow the following program.

- Week 1-2: Modes of stochastic convergence, stochastic orders and uniform integrability.
- Week 3: Weak, strong and triangular arrays laws of large numbers.
- Week 4: I.i.d. and triangular arrays central limit theorems.
- Week 5: Delta method and asymptotic linearity.
- Week 6: Asymptotics of empirical CDFs and quantiles.
- Week 7: Uniform convergence tools.
- Week 8-9: M-estimation, Z-estimation and likelihood asymptotics.
- Week 10-11: Minimax theory for nonparametric regression.
- Week 12: Introduction to semiparametric theory.
- Week 13: Project presentation.

2 Course Assessment

Means of Assessment: Assignments: 20% + Project: 50% + Oral presentation: 30% (in class during the last week of the course). The details and guidelines of the project will be provided as the course progresses.

The grade of the assignment will be based on participation: provided that a student attempts to solve an exercise on the board, the totality (20%) of the assignment grade will be attributed.

3 Course Policies

Rules and Regulations

- The purpose of the Policy on Assessment of Student Learning (PASL) is to provide a set of common principles to guide the assessment of students' learning. Per PASL, assessment should be equitable and consistent, and promote effective learning experiences, a healthy learning environment, and academic integrity. Learn more on the For Students page of the PASL website. Also, see Faculty of Science-specific rules on the implementation of PASL.
- Legally mandated academic accommodations are handled by Student Accessibility and Achievement. For more information see https://www.mcgill.ca/access-achieve/

- In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French written work that is to be graded. This does not apply to courses in which acquiring proficiency in a language is one of the objectives. (Approved by Senate on 21 January 2009)
  Conformément à la Charte des droits de l'étudiant de l'Université McGill, chaque étudiant a le droit de soumettre en français ou en anglais tout travail écrit devant être noté, sauf dans le cas des cours dont l'un des objets est la maîtrise d'une langue. (Énoncé approuvé par le Sénat le 21 janvier 2009)
- McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism, and other academic offenses under the Code of Student Conduct and Disciplinary Procedures (Approved by Senate on 29 January 2003) (See McGill's guide to academic honesty for more information).
- In the event of extraordinary circumstances beyond the University's control, the content and/or assessment tasks in this course are subject to change and students will be advised of the change.

Shortlist of Resources

The Department of Mathematics and Statistics, as well as the University at large, have many resources to help you succeed in this course and throughout your degree. A more extensive list of resources can be found on the Department's EOSW webpage: `https://www.mcgill.ca/mathstat/equity-outreach-wellness`. A shortlist of the most commonly used resources is available here:

- The Wellness Hub is a centralized website for student physical and mental health resources.
- The Math Help Desk is staffed by knowledgeable math students who can help answer your questions related to your courses.
- Statistics Online for Students is a resource that offers online help on statistics courses in the department.
- SUMS is the Society of Undergraduate Mathematics Students. Join their Facebook group to get on their listserv, connect with other students in the departments, and participate in some of their activities/social events.
- GSAMS is the Graduate Student Association for Mathematics and Statistics. All graduate students are represented by GSAMS.
- Advising is an important resource to guide you throughout your degree, and advisors can help with answering questions related to your degree program. Check out the Department's Advising Website to find out how to get in touch with a Departmental Advisor, should you need to do so.

Additional Statements

- As the instructor of this course, I will strive to provide an inclusive learning environment. However, if you experience barriers to learning in this course, please do not hesitate to discuss them with me and/or Student Accessibility and Achievement.
- If you have questions, regarding the course or other areas that you feel I could be of any help, please do not hesitate to contact me at `mehdi.dagdoug@mcgill.ca` or talk to me directly about it. It is my role to help in the best possible way.